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Standards Alignment Technical Documentation

Antikythera Project · MinK Protocol · v3.2

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Executive Summary

This document provides comprehensive technical specifications for implementing a deterministic, auditable standards alignment system. The methodology uses three independent similarity measures to align Student Learning Outcomes (SLOs) from Texas higher education courses (ACGM/WECM corpus) to external frameworks. The framework is rubric-agnostic and works with any external standard including professional licensure (e.g., USCG, DOT), workforce competency frameworks (e.g., NCCER, NAPTA), safety certifications (e.g., BCSP), and industry credentials (e.g., SACA).

Key Characteristics

- Deterministic: Identical inputs produce identical outputs regardless of executor
- Auditable: Complete provenance tracking from raw data to final matches
- Reproducible: Version-controlled code and data with documented dependencies
- Scalable: Standardized file structure supports multiple concurrent projects
- Framework-Agnostic: Works with any external standard or competency framework
- Parsimonious: Each measure runs independently; no score combination or blending

Two-Product Delivery Model

Every standards alignment project produces two deliverables designed for different audiences and purposes:

Product 1: Administrative Report (Excel Workbook)

- Four-tab comprehensive analysis for institutional decision-making
- Coverage Matrix: Visual heatmap showing which courses cover which competencies
- Detailed Matches: Complete data with all similarity scores for drill-down analysis
- Gap Analysis: Unmatched competencies requiring curriculum attention
- Summary Dashboard: Statistics, visualizations, and executive narrative
- Audience: Deans, provosts, curriculum committees, accreditation officers

Product 2: SME Validation Form (Electronic Submission)

- Auto-generated presentation queue for three-person SME panel review
- Matches presented in ranked order by measure, one at a time
- Binary Accept / Not Accept decision per match
- Purpose: Validate match quality through expert judgment with complete audit trail
- Audience: Faculty with discipline expertise (three reviewers required)

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1. System Architecture

1.1 Three-Measure Matching Pipeline

The system runs three independent similarity measures in parallel. Each measure produces its own ranked list of course-to-standard pairs. There is no score combination, cascading filter, or blended confidence score. Scores from each measure are ordinal evidence only — rankings within a measure are meaningful; comparisons across measures are not.

Stage	Method	Text Preprocessing	Output
Stage 1	Sentence Transformers (all-mpnet-base-v2)	Full natural language — no preprocessing	Full corpus ranked by cosine similarity; top-K presented
Stage 2	TF-IDF Vectorization	N-grams (bigrams/trigrams); IDF trained on ACGM+WECM	Full corpus ranked by cosine similarity; top-K presented
Stage 3	Jaccard Similarity	Stop word removal → stemmed tokens (Porter Stemmer)	Full corpus ranked by Jaccard coefficient; top-K presented

Each measure produces a complete ranked list of every course–standard pair in the corpus, sorted by score descending. The user specifies K before execution; the system presents the top-K pairs per measure to the SME panel. There are no admissibility floors and no score-based gates. Every pair is ranked; the panel decides acceptance.

1.2 Design Principles

- Determinism: No random seeds, no probabilistic generation, no model inference variation
- Independence: Each measure runs on its own; scores are never combined or averaged
- Parsimony: Minimal preprocessing preserves interpretability and defensibility
- Stage-Appropriate Processing: Each measure uses text preparation suited to its method
- Framework Agnostic: Method works with any external standard without modification

- Course-Level Analysis: Full SLO text matched as written in official catalog
- Ordinal Integrity: Scores treated as ranking evidence only, not interval measurements
- User-Controlled Queue Depth: K (pairs per measure) and acceptance_target (pairs to accept) are set by the operator before each run

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2. Standardized File Structure

2.1 Purpose and Usage

This section defines the standard directory structure used for ALL standards alignment projects. This structure provides consistency across projects, clear separation of raw data and final outputs, a complete audit trail from source to deliverable, and scalability for multiple concurrent projects.

What You Need to Do: When starting a new project, create this directory structure and populate it according to the specifications in Section 2.3. Maintaining this structure ensures reproducibility and auditability.

2.2 Root Directory Organization

The directory tree below shows the complete structure. Following this layout is REQUIRED:

```
standards-alignment-projects/ ├── _archive/          # Completed projects ├──
_templates/        # Reusable code/configs ├── _shared_data/ │   └──
texas_corpus/ │       ├── acgm_wecm_consolidated.csv │       └──
corpus_statistics.json └── projects/     └── YYYYMMDD_ClientName_Framework/
├── 00_data_prep/         ├── 01_input/         │   ├── source_data/         │
└── provenance/         ├── 02_processed/         ├── 03_interim/        ├──
04_output/         ├── 05_documentation/         ├── 06_code/        └──
07_logs/
```

2.3 Directory Specifications

00_data_prep/

Purpose: Data transformation scripts. Typical Contents: 01_build_texas_corpus.py, 02_catalog_hydration.py, 03_text_standardization.py, 04_deduplication.py, transformation_log.json. Rule: Rule-based transformations only. Document all pragmatic decisions.

01_input/source_data/

Purpose: Raw unmodified data. Typical Contents: acgm_corpus_YYYYMMDD.csv, wecm_corpus_YYYYMMDD.csv, client_catalog_YYYYMMDD.csv,

target_degree_plans_YYYYMMDD.csv, target_standards_YYYYMMDD.csv. Rule: NEVER modify. Treat as read-only archive.

01_input/provenance/

Purpose: Source documentation. Typical Contents: acgm_retrieval_notes.txt, wecm_retrieval_notes.txt, client_data_receipt.txt, target_standards_source.txt. Rule: Document retrieval date, source, and contact for EVERY dataset.

02_processed/

Purpose: Cleaned, validated data. Typical Contents: courses_in_scope.csv, slos_hydrated.csv, slos_cleaned.csv, standards_cleaned.csv, validation_report.txt. Rule: All processing documented in transformation_log.json.

03_interim/

Purpose: Stage outputs. Typical Contents: stage1_sbert_ranked.csv, stage2_tfidf_ranked.csv, stage3_jaccard_ranked.csv, embeddings/. Rule: Complete audit trail. Never delete interim files.

04_output/

Purpose: Final deliverables. Typical Contents: administrative_report_YYYYMMDD.xlsx, sme_presentation_queue_YYYYMMDD.csv, gap_analysis.xlsx. Rule: Timestamped and version-controlled.

05_documentation/

Purpose: Project narratives. Typical Contents: project_brief.md, methodology_decisions.md, sme_feedback.md, lessons_learned.md. Rule: Update throughout lifecycle. Critical for institutional knowledge.

06_code/

Purpose: Analysis scripts. Typical Contents: 00_config.py, 01_preprocessing.py, 02_matching_pipeline.py, 03_export_results.py, requirements.txt. Rule: Follow standard template structure from _templates/.

07_logs/

Purpose: Execution logs. Typical Contents: YYYYMMDD_HHMMSS_preprocessing.log, YYYYMMDD_HHMMSS_matching.log, YYYYMMDD_HHMMSS_export.log, error_log.txt. Rule: Include timestamps, parameters, checksums, and record counts.

3. Data Sources & Preparation

3.1 Required Input Files

For standards alignment projects, five input files are required:

- ACGM Corpus: Texas lower-division academic course catalog with official SLOs
- WECM Corpus: Texas workforce/technical education course catalog with official SLOs
- Client College Catalog: Institution-specific catalog (typically descriptions, not detailed SLOs)
- Target Degree Plans: List of course IDs in target degree programs
- Target Standards: External framework (USCG maritime, DOT transportation, NCCER construction, BCSP safety, NAPTA administrative, SACA systems administration)

3.2 Data Preparation Workflow

1. Filter client catalog by degree plans to identify courses in scope
2. Cross-reference course IDs with ACGM/WECM to extract full official SLOs
3. Concatenate course description with SLOs (descriptions provide context)
4. Clean and standardize text (minimal preprocessing)
5. Validate data quality (nulls, duplicates, encoding)
6. Generate processed datasets ready for matching

> *IMPORTANT: Client catalogs typically lack detailed SLOs. SLO text must be retrieved from the official ACGM/WECM corpus by cross-referencing course identifiers.*

3.3 Consolidated ACGM+WECM Corpus

A consolidated corpus combining all ACGM and WECM courses is maintained for TF-IDF IDF calculation in Stage 2. This corpus should include all ACGM and WECM courses regardless of current project inclusion, course metadata (course_id, title, description), all SLOs for each course, and concatenated text (description + SLOs) for IDF training. IDF should reflect term prevalence across ALL Texas higher education, not just courses in the current project. Update annually when THECB releases updated versions.

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4. Text Preprocessing Specifications

Each measure uses text preprocessing appropriate to its similarity method. This stage-specific approach balances accuracy with interpretability. Preprocessing choices are fixed and must not vary between runs.

4.1 Stage 1: Sentence Transformer

Preprocessing: NONE — Full natural language as-is

Model: all-mpnet-base-v2. Rationale: The sentence transformer is pre-trained on natural language and handles morphology, synonyms, and context internally. Preprocessing degrades performance. All course–standard pairs are scored and ranked; the full ranking is truncated to top-K before SME presentation.

> *Example: "Students will analyze welding symbols and interpret blueprint specifications." → submitted as-is.*

4.2 Stage 2: TF-IDF

Preprocessing: Stop word removal → Lemmatization → WordNet verb synonym expansion → N-grams (1–3)

IDF Training: Consolidated ACGM+WECM (NOT including target standards). All pairs are scored and ranked; the full ranking is truncated to top-K before SME presentation.

Preprocessing pipeline (applied identically to course SLOs and target standards before vectorization):

- Lowercase all text
- Remove English stop words (NLTK stop word list)
- Lemmatize all tokens using NLTK WordNetLemmatizer with POS tagging — verbs are reduced to their base form regardless of tense or conjugation (e.g., "applied," "applying," "applies" → "apply")
- Expand verb lemmas using WordNet synsets: for each verb token, add its synonym lemmas from all synsets to the token stream. Synonym expansion is applied to verbs only; expanding nouns would introduce unacceptable noise in technical domain text
- Build n-grams from the expanded token stream: ngram_range (1, 3)

> *Lemmatization vs. stemming: stemming chops suffixes mechanically ("operating" → "operat"). Lemmatization returns a real dictionary word ("operating" → "operate"), which is required for accurate WordNet synonym lookup. This is why TF-IDF uses lemmatization while Jaccard retains Porter stemming.*

> *Why verbs only for synonym expansion: verbs carry the competency action ("demonstrate," "apply," "analyze"). Expanding technical nouns ("welding" → {joining, bonding, fusion...}) would corrupt the domain-specific IDF signal. Verb expansion aligns learning outcome language across standards that use different action verbs for the same competency.*

TF-IDF vectorizer configuration: max_features 5000; min_df 2; stop_words None (stop words removed in preprocessing step above, not by the vectorizer, so only the pre-cleaned token stream is vectorized).

4.3 Stage 3: Jaccard

Preprocessing: Stop word removal → Stemming (Porter)

Process: lowercase text; remove English stop words (NLTK stop word list); split on whitespace; apply Porter stemmer; calculate |intersection| / |union| of stemmed token sets. Rationale: Stop

word removal prevents common words (“the,” “will,” “students”) from inflating the intersection and producing false high-similarity scores. Stemming normalizes morphological variants (welding = welded = welds). All pairs are scored and ranked; the full ranking is truncated to top-K before SME presentation.

> *Example: Text 1: “Students will perform welding operations” / Text 2: “Applicants shall conduct welded fabrication”. After stop word removal and stemming: {perform, weld, oper} vs {conduct, weld, fabric}. Intersection: {weld} = 1. Union: 5 tokens. Jaccard: 1/5 = 0.20. Without stop word removal, “will” and “shall” would appear in the union and dilute the score while adding no signal.*

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5. Data Processing Workflow

Complete workflow from data receipt to final deliverables consists of four phases.

5.1 Phase 1: Data Ingestion & Validation

Data Receipt

- Receive files from client or retrieve from THECB
- Copy to 01_input/source_data/ with original filenames
- Document source, date, and contact in provenance/
- Calculate and record file checksums (SHA256)

Initial Validation

- Verify file formats and required columns
- Check encoding (UTF-8 required)
- Identify data quality issues
- Confirm degree plan courses exist in catalog

Metadata Capture

- Create project_manifest.json
- Document client requirements
- Identify target framework and set session parameters: K (queue depth per measure) and acceptance_target (number of accepted pairs to terminate review)
- Set project parameters and review scope

5.2 Phase 2: Data Preparation & Cleaning

Catalog Hydration

- Filter client catalog by degree plan course IDs
- Cross-reference with ACGM/WECM for full SLO text

- Concatenate course description with SLOs
- Validate all courses have ACGM/WECM matches
- Save to 02_processed/courses_in_scope.csv

Text Standardization

- Normalize whitespace
- Preserve original text for display
- Remove HTML tags if present
- Check for null/empty values
- Save to 02_processed/slos_cleaned.csv

Target Standards Prep

- Validate target standards file format
- Apply same text cleaning as course SLOs
- Check for duplicate standard IDs
- Save to 02_processed/standards_cleaned.csv

5.3 Phase 3: Three-Measure Matching

Stage 1: Sentence Transformer

- Load all-mpnet-base-v2
- Set deterministic mode: torch.manual_seed(42)
- Generate embeddings for course SLOs (cache in 03_interim/embeddings/)
- Generate embeddings for target standards
- Calculate cosine similarity matrix
- Score all course–standard pairs; sort descending by cosine similarity
- Truncate to top-K pairs (K set by operator in project config)
- Save stage1_sbert_ranked.csv with rank position and score

Stage 2: TF-IDF

- Load consolidated ACGM+WECM for IDF training
- Apply preprocessing pipeline to all texts: lowercase → stop word removal → lemmatize with POS tagging → expand verb synsets via WordNet → build n-grams (1, 3)
- Fit TF-IDF vectorizer (max_features 5000; min_df 2; stop_words None) on preprocessed ACGM+WECM corpus
- Transform preprocessed course SLOs to TF-IDF vectors
- Transform preprocessed target standards to TF-IDF vectors
- Calculate cosine similarity for all pairs
- Score all course–standard pairs; sort descending by cosine similarity
- Truncate to top-K pairs (same K as Stage 1)
- Save stage2_tfidf_ranked.csv with rank position and score

Stage 3: Jaccard

- Apply preprocessing to all texts: lowercase → remove English stop words → split on whitespace → apply Porter stemmer
- Calculate Jaccard coefficient for all pairs: |intersection| / |union| of stemmed token sets
- Score all course–standard pairs; sort descending by Jaccard coefficient
- Truncate to top-K pairs (same K as Stages 1 and 2)
- Save stage3_jaccard_ranked.csv with rank position and score

5.4 Phase 4: Results Compilation & Export

Administrative Report

- Create Coverage Matrix (standards × courses) from accepted SME decisions
- Generate Detailed Matches table with all three scores for accepted pairs
- Compile Gap Analysis with standards that received no Accept decision
- Build Summary Dashboard with statistics and visualizations
- Export to 04_output/administrative_report_YYYYMMDD.xlsx

SME Presentation Queue

- Merge three ranked lists into sequential presentation queue per Section 8 protocol
- Export sme_presentation_queue_YYYYMMDD.csv with measure, rank, score, SLO text, standard text

Documentation

- Update project_manifest.json with execution metadata
- Archive logs to 07_logs/ with timestamps
- Document methodology decisions in 05_documentation/

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6. Technical Specifications

6.1 Required Python Libraries

Install with: pip install -r requirements.txt

Library	Version	Purpose
sentence-transformers	≥2.2.0	Semantic embeddings (all-mpnet-base-v2)
scikit-learn	≥1.3.0	TF-IDF, cosine similarity
pandas	≥2.0.0	Data manipulation

Library	Version	Purpose
numpy	≥1.24.0	Numerical operations
openpyxl	≥3.1.0	Excel export
torch	≥2.0.0	Transformer backend
nltk	≥3.8.0	Tokenization, Porter stemming, WordNetLemmatizer, POS tagging, stop words, WordNet synsets
python-dotenv	≥1.0.0	Configuration management

6.2 Model Specifications

Sentence Transformer:

- Model: all-mpnet-base-v2
- Embedding Dimension: 768
- Max Sequence Length: 512 tokens
- Similarity Metric: Cosine
- Determinism: torch.manual_seed(42)

TF-IDF Preprocessing Pipeline:

- Stop words: NLTK English stop word list (removed before vectorization)
- Lemmatizer: NLTK WordNetLemmatizer with POS tagging (penn_to_wn mapping)
- Verb synonym expansion: WordNet synsets, verbs only; all synonym lemmas appended to token stream
- N-gram Range: (1, 3) applied to the expanded lemma stream
- Max Features: 5000
- Min DF: 2
- Vectorizer stop_words: None (stop words handled upstream)
- Training Corpus: ACGM+WECM consolidated (preprocessed identically before IDF fit)

Jaccard Preprocessing Pipeline:

- Stop words: NLTK English stop word list (removed before stemming)
- Stemmer: Porter Stemmer (NLTK)
- Token Boundary: Whitespace split after lowercase and stop word removal
- Score: |intersection| / |union| of stemmed token sets

6.3 Transformation Log JSON Schema

Required schema for 00_data_prep/transformation_log.json:

```
{   "project_id": "20260201_Miranda_USCG_Maritime",   "session_parameters": {
"K": 30,     "acceptance_target": 3   },   "transformations": [{
"step_number": 1,     "timestamp": "2026-02-01T10:30:00Z",     "script":
```

```
"01_build_texas_corpus.py",     "operation": "consolidate_acgm_wecm",
"input_files": ["acgm.csv", "wecm.csv"],     "output_files":
["acgm_wecm_consolidated.csv"],     "records_in": 15247,     "records_out":
15247,     "transformation_rules": [       "Concatenated description with
SLOs",       "Normalized whitespace",       "UTF-8 encoding enforced"     ],
"quality_flags": [],     "analyst": "Your Name"   }] }
```


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7. Final Deliverables

Every project produces two deliverables: an administrative report for institutional decisions and an SME presentation queue for expert review.

7.1 Product 1: Administrative Report (Excel)

Four-tab Excel workbook (.xlsx):

- Tab 1 — Coverage Matrix: Visual heatmap. Rows = standards, Columns = courses, Cells = Accepted / Not Accepted / Not Reviewed. Summary row/column for coverage counts.
- Tab 2 — Detailed Matches: All SME-accepted pairs. Columns: course_id, course_title, course_slo, standard_id, standard_text, accepted_via_measure, sbert_score, tfidf_score, jaccard_score, review_date, reviewer_panel. Filterable and sortable.
- Tab 3 — Gap Analysis: Standards with no Accepted pair. Columns: standard_id, standard_text, best_sbert_score, best_tfidf_score, best_jaccard_score, notes. Sorted by highest evidence score. For curriculum development.
- Tab 4 — Summary Dashboard: Executive summary (3–5 sentences), coverage statistics (accepted pairs per standard), distribution charts, total courses, total standards, review date, methodology overview, and recommendations.

7.2 Product 2: SME Presentation Queue

Auto-generated CSV and/or electronic form presenting ranked pairs to a three-person SME panel per the protocol defined in Section 8. Each row in the queue contains:

- Queue position (sequential presentation order)
- Measure: SBERT | TF-IDF | Jaccard
- Rank within measure (1 = highest score)
- Score (raw similarity value for reference)
- Course ID and Title
- Course SLO full text
- Target Standard ID and full text
- Decision field: ACCEPT | NOT ACCEPT (populated during review)
- Reviewer notes field (optional)
- Review timestamp (auto-populated)

8. SME Review Protocol

8.0 Session Parameters

Before the matching pipeline runs, the operator sets two parameters that govern the entire review session. These are recorded in the project manifest and must not change mid-session.

Parameter	Type	Definition	Example
K	Integer ≥ 1	Maximum number of pairs per measure included in the presentation queue. The pipeline ranks all corpus pairs by score and truncates each ranked list to K entries. The same K applies to SBERT, TF-IDF, and Jaccard.	K = 20 → up to 60 total pairs in queue (20 per measure)
acceptance_target	Integer ≥ 1	The session terminates automatically as soon as the panel has voted ACCEPT on this many pairs in total, across all measures. This is the primary stopping condition.	acceptance_target = 5 → session ends after 5th ACCEPT

Guidance on setting K: K should be large enough to give the panel meaningful choices but small enough to keep sessions feasible. A starting point for most projects is K = 20–50. If acceptance_target is small relative to K, review will terminate long before the queue is exhausted. If acceptance_target exceeds the number of ACCEPTs the panel produces across all 3K pairs, the session ends when the queue is exhausted.

Guidance on setting acceptance_target: Acceptance_target is the number of aligned pairs you need as evidence of coverage for a given standard. For accreditation contexts, institutions typically require 1–3 accepted alignments per standard competency. Set acceptance_target to match your evidentiary requirement before the session begins.

8.1 Purpose

Subject Matter Expert review validates match quality through expert judgment and produces the final set of accepted course–standard alignments. The protocol is designed to be fast, transparent, and fully auditable. Every decision is recorded with reviewer identity, timestamp, and the measure that sourced the presented pair.

8.2 Panel Composition

Each review session requires exactly three Subject Matter Experts with discipline expertise relevant to the target standard framework. All three reviewers must be present for the session. A simple majority (2 of 3) determines the panel decision for each presented pair.

Requirement	Specification
Panel size	Exactly 3 SMEs per session
Decision rule	Majority vote: 2 of 3 = panel decision
Decision options	ACCEPT or NOT ACCEPT only — no partial credit, no Likert scale
Session timing	After automated matching, before final report delivery
Session parameters	K and acceptance_target confirmed and logged before first pair is presented
Documentation	Each reviewer records their individual vote and name; majority outcome recorded

8.3 Presentation Order Protocol

Pairs are presented to the panel one at a time in the following sequence. The key rule: within a measure, continue presenting next-ranked pairs as long as the panel votes ACCEPT. A NOT ACCEPT decision advances to the next measure in the cycle. After all three measures have cycled, the sequence repeats with the next unreviewed pair in each measure.

The cycle order is fixed: SBERT → TF-IDF → Jaccard → (repeat)

SBERT Rank 1	→	If ACCEPT	→	SBERT Rank 2, 3, 4...	Continue until Not Accept
If NOT ACCEPT ↓					
TF-IDF Rank 1	→	If ACCEPT	→	TF-IDF Rank 2, 3, 4...	Continue until Not Accept
If NOT ACCEPT ↓					
Jaccard Rank 1	→	If ACCEPT	→	Jaccard Rank 2, 3, 4...	Continue until Not Accept
If NOT ACCEPT ↓					
↑ Cycle repeats: next unreviewed SBERT → TF-IDF → Jaccard					

The logic in plain language: Start with the highest-scoring SBERT pair. If the panel accepts it, immediately present the next SBERT pair — and increment the accepted count. Keep going through SBERT until a NOT ACCEPT, then switch to the top unreviewed TF-IDF pair. Same rule applies within TF-IDF. After a NOT ACCEPT in TF-IDF, move to Jaccard. After a NOT ACCEPT in Jaccard, return to the next unreviewed SBERT pair. This cycle repeats until (a) the accepted count reaches acceptance_target, (b) all pairs in all three top-K lists have been reviewed, or (c) the panel adjourns. The accepted count is displayed to the panel at all times.

8.4 Step-by-Step Review Procedure

7. Open the SME Presentation Queue in the review interface (Excel, Forms, or purpose-built tool).
8. All three reviewers confirm their identity in the session log.
9. The facilitator confirms K and acceptance_target aloud and records them in the session log. The running accepted count is displayed prominently throughout.
10. The facilitator presents the first item in the queue: reads the Course SLO and Standard text aloud or displays on screen.
11. Each reviewer records their individual vote: ACCEPT or NOT ACCEPT.
12. The majority outcome is recorded as the panel decision.
13. If ACCEPT: increment the accepted count. If accepted count equals acceptance_target, the session terminates immediately. Otherwise, load the next item from the same measure queue.
14. If NOT ACCEPT: load the next item from the following measure queue (per cycle above).
15. Repeat until acceptance_target is reached, all top-K pairs in all three lists are exhausted, or the panel adjourns. Record the stopping condition and final queue position in session notes.

8.5 What the Panel Reviews

For each presented pair, the panel sees:

- Course ID and Title
- Full Course SLO text
- Full Target Standard text
- Presenting measure and rank position (for transparency)
- Raw score (for reference only — not a recommendation)

The panel's sole question is: Does this course SLO adequately prepare a student to demonstrate this standard? The score is context, not instruction. Panel members should trust their expert judgment over the algorithm's rank.

8.6 Stopping Rules

The review session terminates under any of the following conditions, in priority order:

- Primary: The accepted count reaches acceptance_target. The session closes automatically and no further pairs are presented.
- Secondary: All top-K pairs in all three ranked lists have been reviewed without reaching acceptance_target. The final accepted count is recorded as-is.
- Tertiary: The panel adjourns due to time or quorum. The session may resume at the next unreviewed queue position in a subsequent session with the same panel.

The stopping condition, final accepted count, and final queue position must be documented in the session log. Any unreviewed pairs default to “Not Reviewed” status in the administrative report.

8.7 Session Documentation

After each review session, the following must be recorded in 05_documentation/sme_feedback.md:

- Session date and duration
- Names and roles of all three reviewers
- K and acceptance_target values used
- Total pairs reviewed and total accepted
- Stopping condition (acceptance_target reached / queue exhausted / adjournment)
- Final queue position (if adjourned)
- Any notable patterns or disagreements
- Any recommendations for adjusting K or acceptance_target for future runs

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9. Quality Assurance Protocols

9.1 Pre-Execution Checklist

- All input files in 01_input/source_data/
- Provenance documentation complete
- File checksums calculated
- Project manifest created
- Config file updated with project paths
- Consolidated ACGM+WECM current
- Python environment activated
- transformation_log.json initialized

9.2 During Execution

- Monitor logs for warnings or errors
- Verify record counts at each stage
- Check interim outputs for nulls
- Ensure embeddings cached correctly
- Verify TF-IDF trained on correct corpus
- Confirm Jaccard using stemmed tokens

9.3 Post-Execution Validation

- Data Integrity: No duplicates, all IDs traceable
- Score Ranges: All scores 0–1
- Queue Depth: Each ranked list contains exactly K entries; no more, no fewer
- Ranked Lists Complete: No missing rank positions
- Output Completeness: All required files in 04_output/

- Documentation: Logs archived, manifest updated, transformation log complete

9.4 Spot Check

Manually review 10–20 pairs from each ranked list before SME session:

- Top-ranked SBERT pairs: Expect clear semantic alignment, natural language similarity
- Top-ranked TF-IDF pairs: Expect shared technical terminology
- Top-ranked Jaccard pairs: Expect significant token overlap after stemming
- If top-ranked pairs feel obviously wrong in any list, investigate preprocessing configuration before proceeding to SME review; consider reducing K to focus panel time on the strongest evidence

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10. Re-matching Previous Projects

When re-running with updated methodology or session parameters, follow this protocol.

10.1 Version Control

- Archive Original: Move to _archive/ with original date. Do not modify.
- Create New Project: Use current date (e.g., 20260301_Client_Framework_v2)
- Link Input Data: Symlink or copy original inputs from archived project
- Update Manifest: Reference original project_id, document reasons for re-run
- Document Changes: Create methodology_changes.md in 05_documentation/

10.2 Required Documentation

In methodology_changes.md:

- Original K and acceptance_target values
- Updated K and/or acceptance_target with rationale
- SME feedback summary motivating changes
- Expected impact on queue size
- Processing changes if any
- Analysis date

10.3 Comparison Analysis

Generate comparison report showing delta in total queue items, new pairs appearing in updated ranked lists, pairs no longer appearing, acceptance rate changes by measure, and coverage percentage changes. Save to 05_documentation/comparison_analysis.xlsx.

10.4 Stakeholder Communication

- Clearly identify updated methodology version in filenames
- Explain improvements: “Based on SME review session, adjusted K and acceptance_target for better signal quality and session efficiency”
- Provide side-by-side comparison of accepted alignments
- Recommend which results to use (typically updated)
- Offer to review specific divergent matches

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11. Appendix

11.1 Glossary

Term	Definition
ACGM	Academic Course Guide Manual — Texas lower-division academic course catalog
acceptance_target	Operator-set integer: the session terminates as soon as this many panel ACCEPT decisions have been recorded across all measures
K	Operator-set integer: the number of top-ranked pairs per measure included in the SME presentation queue; same value applies to all three measures
Lemmatization	NLP process that reduces a word to its dictionary base form (lemma) using grammatical context. Tense-aware: “applied,” “applying,” “applies” all return “apply.” Used in TF-IDF preprocessing; required for accurate WordNet lookup
Porter Stemmer	Rule-based suffix-stripping algorithm that reduces words to a stem (not necessarily a real word). Faster than lemmatization; used in Jaccard preprocessing
Stop Words	Common function words (the, will, shall, students, a, of) removed before similarity calculation to prevent them from inflating or diluting scores. NLTK English stop word list used for both TF-IDF and Jaccard
WordNet	NLTK lexical database grouping English words into synsets (synonym sets) with semantic relationships. Used in TF-IDF preprocessing to expand verb tokens with their synonyms, enabling “demonstrate” to match “show” or “exhibit”
Admissibility Floor	Minimum similarity score required for a pair to enter the ranked queue (not an acceptance threshold)
BCSP	Board of Certified Safety Professionals
DOT	Department of Transportation — federal standards
IDF	Inverse Document Frequency — TF-IDF component weighting rare vs. common terms
Jaccard	Similarity measure: |intersection| / |union| of stemmed token sets
NAPTA	National Association Professional Training & Accreditation

Term	Definition
NCCER	National Center for Construction Education and Research
SACA	Systems Administration Credentialing Authority
SBERT	Sentence-BERT — sentence transformer model producing dense semantic embeddings
SLO	Student Learning Outcome
SME	Subject Matter Expert
TF-IDF	Term Frequency–Inverse Document Frequency — lexical similarity measure
USCG	United States Coast Guard — maritime licensing
WECM	Workforce Education Course Manual — Texas technical/workforce course catalog

11.2 References

See Comprehensive_Critique_of_Curriculum_Alignment_Methodology.pdf for full literature review (90+ sources). See also: Text_Similarity_Threshold_Consensus_Across_Methods_and_Applications.md for threshold calibration background and empirical ranges.

- Sentence Transformers: https://www.sbert.net/
- Scikit-learn TF-IDF: https://scikit-learn.org/stable/modules/feature_extraction.html
- NLTK Stemming: https://www.nltk.org/howto/stem.html
- SACSCOC: https://sacscoc.org/
- Texas HECB: https://www.highered.texas.gov/

11.3 Document Control

Version	Date	Changes
1.0	2025-10-01	Initial production release
2.0	2026-01-06	Five-tier confidence system, stage-specific preprocessing, two-product model
3.0 MVP	2026-02-18	Removed confidence tiers and combined scores. Independent ranked lists per measure. SME review replaced with 3-person Accept/Not Accept panel using sequential measure-cycling presentation protocol. Document rebranded as Abulafia under the Antikythera project tree; MinK named as the portable SME review protocol.
3.1 MVP	2026-02-18	Removed admissibility floors entirely. Added user-configurable K (queue depth per measure) and acceptance_target (pairs to accept before session terminates). Both parameters recorded in project manifest and session log.

Version	Date	Changes
3.2 MVP	2026-02-18	TF-IDF preprocessing upgraded: stop word removal, lemmatization (WordNetLemmatizer with POS tagging), and verb-only WordNet synonym expansion before n-gram vectorization. Jaccard preprocessing upgraded: stop word removal (NLTK English list) before Porter stemming.

Prepared by: _____ Date: _____

Reviewed by: _____ Date: _____

Approved by: _____ Date: _____